

Experiment 3-Low Wind: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	2000	1.031	0.722	1.806	67.7%	93.1%	97.6%	8.039	12.463	3.647	7.182	0.301	0.459	0.572
Proposed	2000	1.291	0.632	1.381	78.8%	94.7%	95.9%	6.331	12.452	4.253	10.362	0.283	0.441	0.536
PPO-GRU	1992	1.553	0.726	1.831	72.0%	90.5%	91.9%	6.461	12.832	4.056	7.827	0.271	0.420	0.599
MPC	1656	2.427	2.528	2.904	0.0%	18.8%	99.6%	1.570	2.543	1.237	2.218	0.136	0.136	0.143

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	2000	100.0%	0	0.0%	0.0%	0	nan	0	0
Proposed	2000	2000	100.0%	0	0.0%	0.0%	0	nan	0	0
PPO-GRU	2000	1992	99.6%	8	0.4%	0.4%	8	100.0%	0	0
MPC	2000	1656	82.8%	344	17.2%	17.2%	344	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	344	344	100.0%	7	2.0%	0.980
Proposed	344	344	100.0%	10	2.9%	1.105
PPO-GRU	344	342	99.4%	22	6.4%	1.379